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4 / 20 submitted

Q1. Arrays

CODE

```
#include<stdio.h>
int main(){
    int production[12] = {5000, 4800, 5100, 4950, 4700, 5100, 5000, 4800, 5000, 5000 ,4900, 4800};
    int target = 5000;
    int total_target= target*12;

    int total_production=0;
    int valid_months=0;

    // printf("Enter Production values for 12 months:\n");
    //production[12]= [5000, 4800,5100,4950,4700,5100,6000,4800,5000,5000,4900,4800];
    for (int i =0;i<12;i++){

        if (production[i]==-1){
            printf("Equipment failure encountered! plant shutodnw\n");

        }
        if (production[i]<0){
            printf("invalid production value, Skipping month %d", i+1);
            continue;
        }
        total_production +=production[i];
        valid_months++;
    }
    if(valid_months>0){
        float avg_monthly = (float)total_production/valid_months;
        float efficiency = ((float)total_production/(target*valid_months))* 100.0;

        printf("\n Performance sumar\n");
        printf("Total Annual Production: %d batteries\n",total_production);
        printf("Average Monthly Production: %.0f batteries\n", avg_monthly);
        printf("production Efficiency: %.2f%%\n", efficiency);

        if(avg_monthly>target){
            printf("Status:performance Incetive awarded. 10%% bonux\n");
        }
        else if(avg_monthly <(0.80*target)){
            printf("Status:Marked for inspection, Below 80%%\n");

        }
        else{
            printf("Status: on track( target met, normal operation)");
        }
    }
    else{
        printf("No valid Production data was processed.\n");
    }
    return 0;
}
```

INPUT

5000 4800 5100 4950 4700 5100 6000 4800 5000 5000 4900 4800

OUTPUT

(no output)

Q2. Arrays

CODE

```
#include<stdio.h>
int main(){
    int quality[20] = {95,82,65,45,-2,91,78,70,55,88,92,60,-1,80,75,90,85,68,50,94};
    int delay[20] = {1,3,4,6,-2,0,5,2,0,4,2,8,-1,1,3,0,2,1,5,0};

    int total_quality=0;
    int on_time_count=0;
    int evaluated_projects=0;

    printf("Project Classisificatoins \n");
    for(int i = 0;i<20; i++){
        if(quality[i]==-1 || delay[i]==-1){
            printf("\nManagement Stopped the evauluation at Project %d.\n",i+1);
            break;
        }if(quality[i]<0|| delay[i]<0){
            printf("project %d:Concelled (skipped)\n", i+1);
            continue;
        }
        total_quality+=quality[i];
        if (delay[i]==0){
            on_time_count++;
        }
        evaluated_projects++;
        printf("project %d (quality;%d, Delay: %d days): ", i+1, quality[i], delay[i]);
        if (quality[i] ≥90 && delay[i]≤2){
            printf("Excellent\n");
        }else if (quality[i]≥75 && quality[i]≤89 && delay[i]≤5) {
            printf("good\n");
        }else if (quality[i]≥60 && quality[i] ≤74){
            printf("average\n");
        }else{
            printf("Needs Improvements\n");
        }
    }
    if(evaluated_projects>0){
        float avg_quality = (float)total_quality/evaluated_projects;
        float delivery_percentage = ((float)on_time_count/evaluated_projects) * 100.0;
        printf("\n      Overage Perfoamnce Symmary      \n");
        printf("total evualted projects: %d\n", evaluated_projects);
        printf("Average Quality Score: %.2f\n", avg_quality);
        printf("On-Time Delivery Percentage ( 0 Days delay): %.2f%%\n", delivery_percentage);

    }else {
        printf("\n No valid projects were evualited.\n");
    }
    return 0;}
```

INPUT

input directly coded into program due to problem with complier

OUTPUT

(no output)

Q3. Arrays

— Not Submitted —

Q4. Arrays

— Not Submitted —

Q5. Arrays

— Not Submitted —

Q6. Arrays

CODE

```
#include <stdio.h>
int main(){
int bad_chip_count[30]= {50,120,250,600,-2,80,150,400,320,90,110,-2,200,50,550,70,130,-1,90,100,200,300,400,500,600,700,800,90
0,100,200};
const int daily=10000;
    int days= 0;
    int grand_accepted=0;
    float sum_defect_rate=0.0f;

    int idx = 0;
    while(idx<30){
        int flaws=bad_chip_count[idx];
        if(flaws===-1){
            printf("half Day %d: machine fault record. Terminator", idx+1);
            break;

        }
        if (flaws===-2){
            printf("skip, day%d Scheduled maintnenace day\n",idx+1);
            idx++;
            continue;
        }
        int valid_chips = daily-flaws;
        float rate = ((float)flaws/ daily) *100.0f;

        grand_accepted+=valid_chips;
        sum_defect_rate+=rate;
        days++;
        printf("Days %02d | Defeccts: %d | Accepted: %d | Rate: %.2f%% | Grade: ", idx+1, flaws,valid_chips,rate);

        if (rate<1.0f){
            printf("Grade a\n" );

        } else if (rate≥1.0f && rate ≤3.0f){
            printf("Grade B\n");
        }else if (rate >3.0f && rate ≤ 5.0f){
            printf("grad c\n");
        }else{
            printf("reject batch\n");
        }
        idx++;
    }
    if (days>0){
        printf("\n—————\n");
        printf("Active Operatinonal Days: %d\n", days);
        printf("Total Good Chips Produced: %d\n", grand_accepted);
        printf("Mean Defect Rate: %.2%\n", sum_defect_rate/days);
        printf("||\\|—————\n");
    }

    return 0;
}
```

OUTPUT

(no output)

Q7. Arrays

CODE

```
#include <stdio.h>

int main(){
    const int Server = 5000;
    int customer_ids[]={101,102,103,104,105,106,107,108};
    int plan_types[]={1,2,3,1,2,3,2,1};
    int storage_used[] = { 96, 450,-1,110,520,1950,480,80};

    int num_cusomters = sizeof(customer_ids)/ sizeof(customer_ids[0]);
    int total_server_used = 0;

    printf("Cloud Storage Subscription Management \n\n");
    for (int i=0;i<num_cusomters;i++){
        if(storage_used[i]<0){
            printf("Customer %d: Account Inactivy shkkinging\n\n",customer_ids[i]);
            continue;
        }
        int plan_limit=0;
        if (plan_types[i]==1){
            plan_limit=100;
        }else if(plan_types[i]==2){
            plan_limit=500;
        }else if (plan_types[i]==3){
            plan_limit=2000;
        }
        int used = storage_used[i];
        int remaining = plan_limit -used;
        int extra_gb;
        int extra_charge = 0;
        if (remaining <0){
            extra_gb=-remaining;
            remaining = 0;
            extra_charge = extra_gb*4;
        }
        total_server_used += used;
        if (total_server_used ≥ Server){
            printf("Critical Error:Server Storage if full! Stopping processing\n");
            break;
        }
        printf("Customer ID: %d\n", customer_ids[i]);
        printf("Plan Limit:%d GB| Used: %d GB | Remaining: %d GB\n", plan_limit,used,remaining);
        if (extra_charge>0){
            printf("Extra Storage Used: %d GB | Extra Charge : Rs. %d\n", extra_gb,extra_charge);
        }else{
            printf("Extra Charge : $0\n");
        }

        float usage_percentage = ((float)used/plan_limit) *100.0;
        if (usage_percentage > 95.0){
            printf("Warning:Storage usage is at %.1f%% (>95%% threshold)!\n", usage_percentage);
        }else {
            printf("Status: Normal (%.1f%% used)\n", usage_percentage);
        }
        printf("\n");
    }
    printf("Total Server Storage Used:%d / %d GB\n", total_server_used, Server);
    return 0;
}
```

OUTPUT

(no output)

Q8. Arrays

— Not Submitted —

Q9. Arrays

— Not Submitted —

Q10. Arrays

— Not Submitted —

Q11. Arrays

— Not Submitted —

Q12. Arrays

— Not Submitted —

Q13. Arrays

— Not Submitted —

Q14. Arrays

— Not Submitted —

Q15. Arrays

— Not Submitted —

Q16. Arrays

— Not Submitted —

Q17. Arrays

— Not Submitted —

Q18. Arrays

— Not Submitted —

Q19. Arrays

— Not Submitted —

Q20. Arrays

— Not Submitted —